

Standard forms

Logic in Computer Science Lab
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Based on prof. Tamás Kádek's lectures

Fixed Occurrences

Definition

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first order language, $A \in Form$ be a formula, and $x \in Var$ be a variable.

- A fixed occurrence of the variable x in the formula A is **free** if it is not in the subformulae $\forall xB$ or $\exists xB$ of the formula A .
- A fixed occurrence of the variable x in the formula A is **bound** if it is not free.

Open and Closed Formulae

Definition

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first order language and $A \in Form$ be a formula.

- If $FreeVar(A) \neq \emptyset$, then the formula A is an **open formula**.
- If $FreeVar(A) = \emptyset$, then the formula A is a **closed formula**.

Remark

- The formula A is open if there is at least one variable which has at least one free occurrence in A .
- The formula A is closed if there is no variable which has a free occurrence in A .

Which are open and which are closed formulae?

1. $\exists z(P(y) \wedge \exists x P(f(x), z) \supset P(x, y))$
2. $\forall x(\exists z Q(x, z) \supset \forall z(Q(c, x) \wedge Q(x, y)))$
3. $\exists x \forall y Q(x, y) \supset \forall y \exists x Q(y, x)$
4. $\exists y \exists x \forall z((R(z, z) \supset y = z) \wedge \forall y(P(g(y, z)) \wedge y = x))$

Exercises

Mark the bounded (indexed) occurrences in the formula

$\neg \forall v (Q(g(u, c), z_1) \wedge (Q(v, v_2) \equiv g(x, z_3) = f(w_4)))$, where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$, $P1=\{P\}$ and $P2=\{Q,R\}$?

Select one or more:

- ☐ a. 4
-  ☒ b. 2
- ☐ c. 3
- ☐ d. 1

Mark the free (indexed) occurrences in the formula

$\exists u_4 \exists x (\exists z \forall x \forall u_2 Q(u, g(w, g(x, u_1))) \supset Q(v, z) \equiv Q(y_3, y))$, where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$, $P1=\{P\}$ and $P2=\{Q,R\}$?

Select one or more:

☐ a. 2

☐ b. 1

☒ c. 3

☐ d. 4

Mark the free (indexed) occurrences in the formula

$v = w_1 \supset \forall u \exists w (R(x_2, f(v)) \vee \exists u \forall x Q(u_3, f(h(w, v_4))))$, where $F(0) = \{c\}$, $F1 = \{f\}$, $F2 = \{g, h\}$, $P1 = \{P\}$ and $P2 = \{Q, R\}$?

Select one or more:

☐ a. 3

 ☒ b. 4

 ☒ c. 2

 ☒ d. 1

Mark the free (indexed) occurrences in the formula

$\exists u_4 \exists x (\exists z \forall x \forall u_2 Q(u, g(w, g(x, u_1))) \supset Q(v, z) \equiv Q(y_3, y))$, where $F(0) = \{c\}$, $F1 = \{f\}$, $F2 = \{g, h\}$, $P1 = \{P\}$ and $P2 = \{Q, R\}$?

Select one or more:

☐ a. 4

☐ b. 2

☐ c. 1

☒ d. 3

Mark the free (indexed) occurrences in the formula

$\forall y (w = g(v, z) \wedge \forall v_3 \forall w_2 P(f(h(g(z, c), w_1))) \equiv R(v, u_4))$, where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g, h\}$, $P1=\{P\}$ and $P2=\{Q, R\}$?

Select one or more:

☐ a. 2

 ☒ b. 4

☐ c. 1

☐ d. 3

Standardized form

Definition 4.30

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first order language and $A \in Form$ be a formula.

The formula A is standardized if

1. $FreeVar(A) \cap BoundVar(A) = \emptyset$;
2. all bound variables of the formula A have exactly one occurrences next a quantifier.

Congruence

Skeleton of the formula:

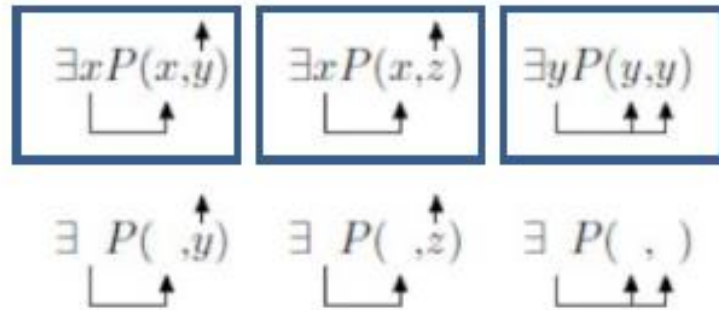
$$\forall x (A(x) \wedge \exists x B(x) \supset C(x)) \vee B(x)$$

$$\forall (A() \wedge \exists B()) \supset C()) \vee B(x)$$

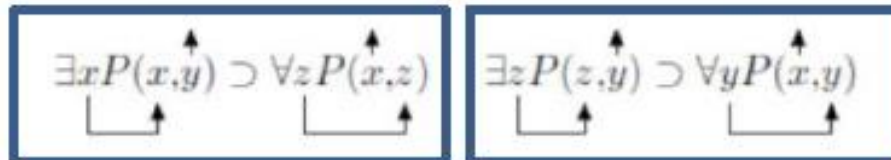
2 formulas are congruent if they have the same skeleton.

Congruence

(a) Not congruent formulae (not variants of themselves) because the



(b) Congruent formulae (variants)



Standard Form — How it looks like

$$\exists y(\forall x P(x) \supset \forall x Q(x, y))$$

The formula (in general form)

$$\exists y(\forall x P(x) \supset \forall w Q(w, y))$$

The formula in Standard Form

$$\exists z(\forall x P(x) \supset \forall w Q(w, z))$$

The formula in Standard Form

$$\exists y(\forall z P(z) \supset \forall x Q(x, y))$$

The formula in Standard Form

Construct the standardized forms of the formulae

1. $\exists z(P(y) \wedge \exists x P(f(x), z) \supset P(x, y))$
2. $\forall x(\exists z Q(x, z) \supset \forall z(Q(c, x) \wedge Q(x, y)))$
3. $\exists x \forall y Q(x, y) \supset \forall y \exists x Q(y, x)$
4. $\exists y \exists x \forall z((R(z, z) \supset y = z) \wedge \forall y(P(g(y, z)) \wedge y = x))$

Construct a standard form of the formula:

$$\forall x(\neg \exists x A(x) \vee B(x) \supset \neg \exists x A(x)) \supset (\neg \forall x B(x) \supset A(x) \vee \neg \exists x A(x))$$

Substitutions

Which are allowed substitutions where A is the formula

$$\neg(x = c \wedge Q(z, c)) \wedge \forall y(d = d \vee P(z))$$

a) $A(z||d)$

This means: a) Can you replace all occurrences of the z variable by d without changing dependencies, the semantical meaning of the formula?

b) $A(z||x)$

c) $A(z||g(y, x))$

d) $A(y||y)$

(A substitution is not allowed, if a free variable will be bound after the substitution or if it changes the structure of the dependency in the formula, so the semantic meaning of the formula.)